

Electrical/Computer Engineer interested in the intersection of control systems, signal processing, and embedded systems.

Education

Master of Science in Computer Engineering (Accelerated Master's Program) Virginia Tech – Software and Machine Intelligence <i>Advisers: Dr.Thinh Doan (UT Austin) and Dr.Michael Hsiao (Virginia Tech)</i>	May 2025 Blacksburg, Virginia
Bachelor of Science in Electrical & Computer Engineering (double major) Virginia Tech – Control Systems and Machine Learning	May 2024 Blacksburg, Virginia

Technical Experience

Virginia Tech · Robotics Research Robotics Researcher - Developing a 6-axis robotic manipulator and an accompanying ROS2–Gazebo & MuJoCo simulation using Gymnasium for deploying custom reinforcement learning algorithms. - Integrating CNN-based object detection (YOLOv8), RNN-based language parsing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks.	Aug 2023 – May 2025 Blacksburg, Virginia
Graduate Teaching Assistant - Teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design. - Assisting with hands-on projects to illustrate and integrate analog and digital filter design and application on breadboards and TI MSP432 development boards.	Aug 2024 – May 2025 Blacksburg, Virginia
Jacobs Space Exploration Group · Mars Ascent Vehicle (MAV) Thrust Vector Control Intern - Developed thrust vector control testing hardware and software for NASA's Active Inertial Load Simulator at the Marshall Space Flight Center. - Created and ran tests to develop a mathematical model of an electro-mechanical actuator – used Python, MATLAB, and LabView. - Derived control algorithms for a load-simulating actuator, in Simulink, to simulate external loads placed on the Mars Ascent Vehicle's thrust vector control actuators during flight. - Designed and integrated IIR filters to remove high frequency noise from a load cell and linear variable differential transformer (LVDT).	May 2024 – Aug 2024 Huntsville, Alabama (Merrit Island, Florida)
Grenoble Electrical Engineering Laboratory · Microgrid Inverters Control Systems Research Intern - Researched inverter control systems – designed to be robust to islanding events and avoid future infrastructure problems on the French power grid. - Simulated neutral point capacitive and balancing topologies using 4-leg inverters in Simulink. Tested PI control, PR control, Clarke and Park Transforms with HIL simulations.	Jun 2023 – Aug 2023 Grenoble, France
Naval Surface Warfare Center (Carderock Division) · Hospital Sea Trains Concept Research Intern Developed concept hospital sea-train designs at the Center for Innovation in Ship Design and estimated fuel consumption and electrical power loads of the concept sea-trains.	Jun 2022 – Aug 2022 West Bethesda, Maryland
Skills	
Software: C/C++, Python, MATLAB, Simulink, GNU/Linux, Git, ROS2, Gazebo, Make, CMake, Labview, Qt, PyTorch, OpenCV, $\LaTeX$, Verilog, FreeRTOS, Autodesk Inventor (Certified), SolidWorks, Rhino	
Hardware: PCB Design and Assembly, Breadboarding, Computer Architecture, Oscilloscope, Multimeter, 3D-Printing	
Projects	
6-Axis Robotic Arm  3D printed robot arm, built using stepper motors and pulleys. - ROS2 Jazzy control and Gazebo Harmonic simulation.	Aug 2024 – Present
LQI Rocket Landing Simulation  - Landing a <i>very simplified</i> simulated rocket in MATLAB using optimal control. - Designed a LQI controller for full-state feedback and setpoint tracking of a landing trajectory.	Aug 2023 – May 2024
Closed Loop Stepper Motor  - Backdrivable stepper motor driver using closed loop control and a magnetic encoder for feedback. 4-layer PCB mounts to the back of the motor with CAN and power connections.	Dec 2023 – May 2024
Design Teams Solar Car & Human Powered Submarine  - Overall E/E architecture of the Solar Car. - Single board computer and LCD to display relevant data to the submarine pilot.	Oct 2020 – Mar 2023